

Hyperbolas



Standard form

- 1 For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:
 - a the vertices
 - b the foci
 - c the equations of the asymptotes
 - 2 Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.
 - 3 Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.
 - a Write the equation in standard form by completing the square.
 - b Identify the conic and state its center.
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Identifying conics

- 4 Identify each conic section from its equation:

- a $x^2 + y^2 = 49$ - c $y^2 - 4x^2 = 16$	- b $\frac{x^2}{9} + \frac{y^2}{25} = 1$ - d $y = 3x^2 - 6x + 1$
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- 5 A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.